Certified reliability for any routing solver: screening buffered plans against travel-time uncertainty

Suhan Ali
Independent researcher, Lahore, Pakistan
E-mail: syedsuhanali9@gmail.com
ORCID: 0009-0004-5079-7684

Corresponding author: Suhan Ali (address as above). The author is an independent researcher with no institutional affiliation, so no institutional e-mail address is available; the ORCID record identifies the author.
Word count: about 11,800 words from the abstract to the end of the appendix, including tables and captions, excluding references.

Acknowledgements. The author thanks the researchers who answered methodological questions by e-mail during the project.
Disclosure of interest. The author reports there are no competing interests to declare.
Funding. This research received no external funding.
Data availability statement. All code, instances, records and analysis scripts are openly available on Zenodo (doi:10.5281/zenodo.23185704, all versions; the analyses in the manuscript are in version v4.1 and later). The Amazon route sample is redistributed under its CC BY-NC 4.0 licence; the New York taxi records are public and are not copied.
Author contributions (CRediT). Suhan Ali: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing – original draft, writing – review and editing.
Use of generative AI. Claude (Anthropic; mainly Claude Opus 5.5, with earlier Claude models during 2026) and DeepSeek (DeepSeek V4.1 Flash 1017) were used to write and debug the simulation and analysis code, to review that code, to run and check the statistical analyses, to search the literature, and to draft and revise the text of the manuscript, including its wording. They were used under their providers' terms of use. The research question, the study design and the confirmatory protocols were the author's. Every experiment was run from the archived code, every reported number is regenerated from the archived records, and the author takes full responsibility for the content.

Biographical note. Suhan Ali is an independent researcher based in Lahore, Pakistan, working on vehicle routing and delivery planning under travel-time uncertainty.